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Кафедра інформаційних технологій

ЗВІТ

з лабораторної роботи № 15

з дисципліни «Основи програмування»

на тему:

«Позиціонування віджетів за допомогою методів place і grid

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Визначення завдання

Для завдання було обрано тему “Калькулятор для виробів з тонколистових металів”. Було реалізовано три команди по заданій специфіці

- Показати доступні тонколистові метали
- Розрахувати розгортку та вартість труби
- Розрахувати приблизну вартість коліна

В якості основи для програми було використано раніше розроблений програмний комплекс в якості індивідуального творчого завдання

Код до завдання

Код UI/gui/TkinterViewer.py

```
import tkinter as tk
from core.events import Event, EventType
from UI.common.presentation.proxy import ParameterUIAssembler
from UI.common.presentation.hint import RangeHint, OrderHint,
ChoiceHint, LargeTextHint, ListboxHint


class TkinterViewer:
    def __init__(self, context):
        self.context = context
        self.root = tk.Tk()
        self.root.title("Individual Creative Assignment")
        self.root.geometry("1000x600")

        self.root.protocol("WM_DELETE_WINDOW", self.on_closing)

        self.services = {"Вийти": ""}
        self.commands = {}

        self.sidebar = tk.Frame(self.root, width=300, bg="#ffffff",
relief="sunken", bd=1)
        self.sidebar.pack(side=tk.LEFT, fill=tk.Y, padx=5, pady=5)
        self.sidebar.pack_propagate(False)

        self.footer = tk.Frame(self.root, height=30, bg="#ffffff",
relief="sunken", bd=1)
        self.footer.pack(side=tk.BOTTOM, fill=tk.X, padx=5, pady=5)

        self.status = tk.Label(self.footer, text="Ready",
fg="black", bg="#ffffff", anchor="w")

        self.status.pack(fill=tk.X)

        self.main_area = tk.Frame(self.root, bg="white")
        self.main_area.pack(side=tk.RIGHT, fill=tk.BOTH,
expand=True, padx=5, pady=5)

        self.current_params_values = {}
        self.current_params_widgets = {}

    def on_closing(self):
        self.context.events.emit(Event(EventType.EXIT))

    def handle_event(self, event):
        self.status.config(text=event)
        if event.type == EventType.START:
```

```
                self.show_services_sidebar()

            elif event.type == EventType.SERVICE_AVAILABLE:
                service_name = event.data["service_name"]
                self.services[service_name] = event.data["service"]
                self.show_services_sidebar()

            elif event.type == EventType.COMMAND_AVAILABLE:
                command_name = event.data['name']
                self.commands[command_name] = {
                    "command_cls": event.data["command_cls"],
                    "service": event.data["service_name"]
                }

            elif event.type == EventType.SHOW_SERVICES_MENU:
                self.show_services_sidebar()

            elif event.type == EventType.SERVICE_SELECTED:
                self.show_commands_sidebar(event.data["service_name"])

            elif event.type == EventType.EXECUTE_COMMAND:
                self.show_params_form(event.data["service_name"],
event.data["command_name"])

            elif event.type == EventType.COMMAND_EXECUTED:
                self.show_result("Результат",
str(event.data["result"]))

            elif event.type == EventType.ERROR:
                self.show_result("Помилка",
str(event.data["error_message"]))

            elif event.type == EventType.EXIT:
                self.root.destroy()

        def clear_sidebar(self):
            for widget in self.sidebar.winfo_children():
                widget.destroy()

        def clear_main_area(self):
            for widget in self.main_area.winfo_children():
                widget.destroy()

        def show_services_sidebar(self):
            self.clear_sidebar()
            tk.Label(self.sidebar, text="Сервіси", font=("Arial", 14,
"bold"), bg="#f0f0f0").pack(pady=10)

            for name in self.services:
                if name == "Вийти":
                    tk.Button(self.sidebar, text=name, width=300,
command=self.on_closing, bg="#ff9999").pack(
                        side=tk.BOTTOM)
                else:
                    tk.Button(self.sidebar, text=name, width=300,
                              command=lambda n=name:
self.context.events.emit(
                                  Event(EventType.SERVICE_SELECTED,
service_name=n))).pack()
```

```
        def show_commands_sidebar(self, service_name):
            self.clear_sidebar()
            tk.Label(self.sidebar, text=f"Команди\n{service_name}",
font=("Arial", 12, "bold"), bg="#f0f0f0").pack(pady=10)

            commands_for_service = [k for k, v in self.commands.items()
if v["service"] == service_name]

            for cmd_name in commands_for_service:
                tk.Button(self.sidebar, text=cmd_name, width=300,
                          command=lambda c=cmd_name:
self.context.events.emit(
                              Event(EventType.EXECUTE_COMMAND,
command_name=c, service_name=service_name)
                          )).pack()

            tk.Button(self.sidebar, text="<- До сервісів", width=300,
command=self.show_services_sidebar,
                      bg="#cccccc").pack(
                side=tk.BOTTOM)

        def show_result(self, title, message, color="white"):
            self.clear_main_area()
            self.main_area.configure(bg=color)
            tk.Label(self.main_area, text=title, font=("Arial", 16,
"bold"), bg=color).pack(pady=20)
            tk.Label(self.main_area, text=message, font=("Arial", 12),
bg=color, justify=tk.LEFT, wraplength=600).pack(
                padx=20, pady=10, fill=tk.BOTH, expand=True)
            tk.Button(self.main_area, text="Очистити", width=20,
command=self.clear_main_area).pack(pady=20)

        def show_params_form(self, service_name, command_name):
            self.clear_main_area()
            self.main_area.configure(bg="white")
            service = self.services[service_name]
            command_cls = self.commands[command_name]["command_cls"]

            tk.Label(self.main_area, text=f"Команда: {command_name}",
font=("Arial", 14, "bold"), bg="white").pack(pady=10)
            if hasattr(command_cls, "description"):
                tk.Label(self.main_area, text=command_cls.description,
wraplength=600, bg="white").pack(pady=5)

            assembler = ParameterUIAssembler(command_cls, service)
            ui_params = assembler()

            if not ui_params:
                self.execute_with_params(command_cls, service, [])
                return

            def get_order(p_proxy):
                for hint in p_proxy.ui_hints:
                    if isinstance(hint, OrderHint):
                        return hint.order_number
                return 999

            ui_params.sort(key=get_order)

            self.current_params_values = {}
```

```
            self.current_params_error_labels = {}

            form_frame = tk.Frame(self.main_area, bg="#cccccc")
            form_frame.pack(fill=tk.BOTH, expand=True, padx=20,
pady=10)

            form_frame.columnconfigure(0, weight=0)
            form_frame.columnconfigure(1, weight=1)
            form_frame.columnconfigure(2, weight=0)

            # Header
            tk.Label(form_frame, text="Параметр", font=("Arial", 10,
"bold"), bg="#eeeeee", relief=tk.RIDGE, bd=1).grid(
                row=0, column=0, sticky="nsew")
            tk.Label(form_frame, text="Значення", font=("Arial", 10,
"bold"), bg="#eeeeee", relief=tk.RIDGE, bd=1).grid(
                row=0, column=1, sticky="nsew")
            tk.Label(form_frame, text="Опис", font=("Arial", 10,
"bold"), bg="#eeeeee", relief=tk.RIDGE, bd=1).grid(
                row=0, column=2, sticky="nsew")

            for i, param_proxy in enumerate(ui_params):
                row_index = (i + 1) * 2 - 1

                tk.Label(form_frame,
text=f"{param_proxy.display_name}:", width=25, anchor="w", bg="white",
relief=tk.RIDGE, bd=1).grid(
                    row=row_index, column=0, sticky="nsew", rowspan=2)

                tk.Label(form_frame, text=f"{param_proxy.description}",
width=25, anchor="w", bg="white", relief=tk.RIDGE, bd=1,
wraplength=200).grid(
                    row=row_index, column=2, sticky="nsew", rowspan=2)

                parse_type = param_proxy.parse

                choices = []
                range_hint = None
                large_text = False
                listbox_hint = None

                for hint in param_proxy.ui_hints:
                    if isinstance(hint, ChoiceHint):
                        choices = hint.choices
                    elif isinstance(hint, RangeHint):
                        range_hint = hint
                    elif isinstance(hint, LargeTextHint):
                        large_text = True
                    elif isinstance(hint, ListboxHint):
                        listbox_hint = hint

                widget_container = tk.Frame(form_frame, bg="white",
relief=tk.RIDGE, bd=1)
                widget_container.grid(row=row_index, column=1,
sticky="nsew")

                if large_text:
                    text_widget = tk.Text(widget_container, height=5,
width=40)
                    text_widget.pack(fill=tk.X, expand=True)
```

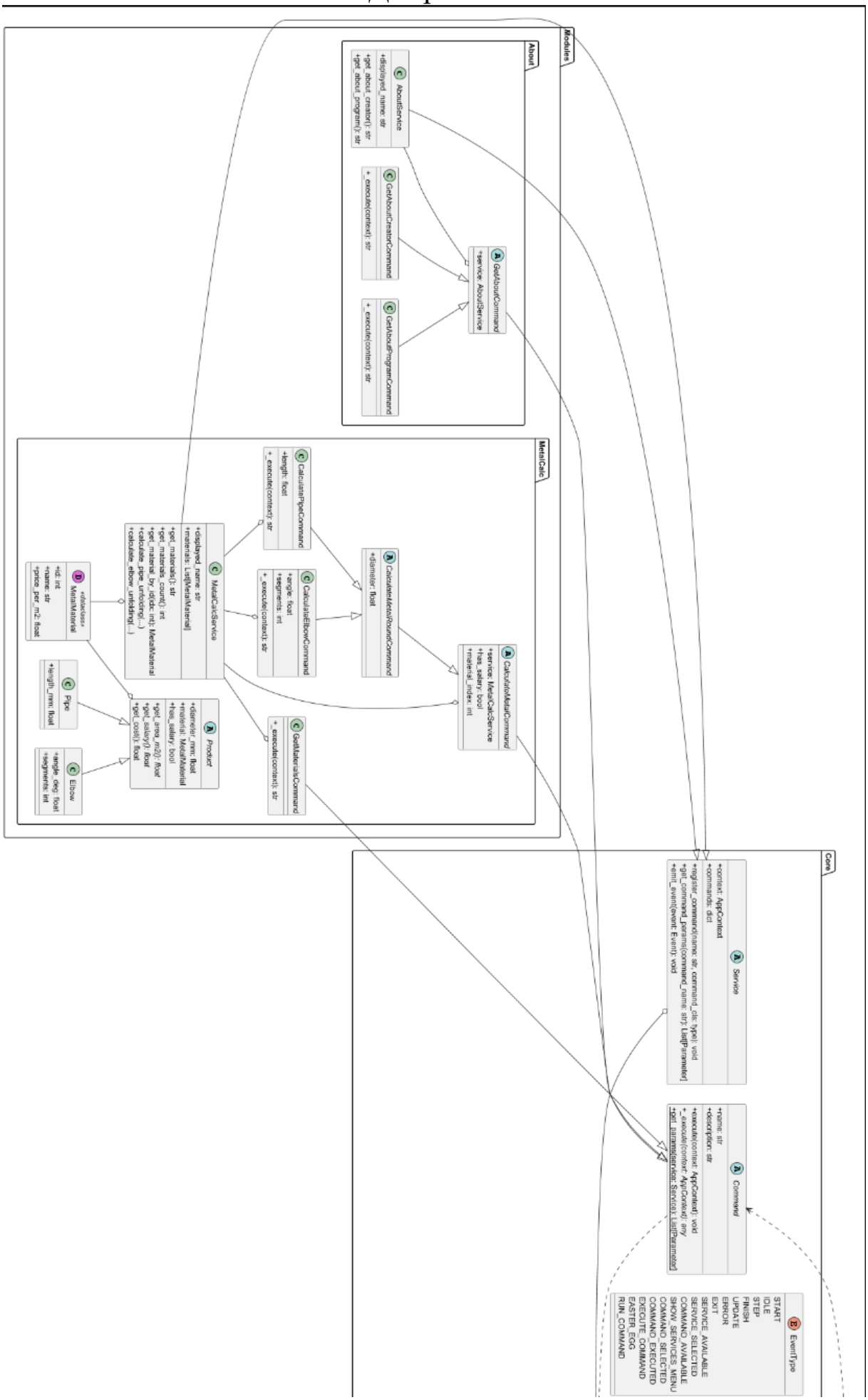
```
                    self.current_params_values[param_proxy.name] =
text_widget

                elif listbox_hint:
                    lb_frame = tk.Frame(widget_container)
                    lb_frame.pack(fill=tk.X, expand=True)
                    scrollbar = tk.Scrollbar(lb_frame,
orient=tk.VERTICAL)
                    listbox = tk.Listbox(lb_frame,
height=listbox_hint.height, yscrollcommand=scrollbar.set)
                    scrollbar.config(command=listbox.yview)
                    scrollbar.pack(side=tk.RIGHT, fill=tk.Y)
                    listbox.pack(side=tk.LEFT, fill=tk.BOTH,
expand=True)
                    for choice in listbox_hint.choices:
                        listbox.insert(tk.END, choice)
                    self.current_params_values[param_proxy.name] =
listbox

                elif choices:
                    val_var = tk.StringVar(value=str(choices[0]))
                    self.current_params_values[param_proxy.name] =
val_var
                    option_menu = tk.OptionMenu(widget_container,
val_var, *choices)
                    option_menu.pack(side=tk.LEFT)

                elif parse_type == bool:
                    var = tk.BooleanVar(value=False)
                    self.current_params_values[param_proxy.name] = var
                    tk.Checkbutton(widget_container, variable=var,
bg="white").pack(side=tk.LEFT)

                elif parse_type in (int, float):
                    val_var = tk.StringVar(value="0")
                    self.current_params_values[param_proxy.name] =
val_var

                    def validate_range(event, var=val_var,
rh=range_hint, pt=parse_type):
                        if rh is None:
                            return
                        try:
                            val = pt(var.get())
                            if rh.min_value is not None and val <
rh.min_value:
                                var.set(str(rh.min_value))
                            elif rh.max_value is not None and val >
rh.max_value:
                                var.set(str(rh.max_value))
                        except ValueError:
                            if rh.min_value is not None:
                                var.set(str(rh.min_value))
                            elif rh.max_value is not None:
                                var.set(str(rh.max_value))
                            else:
                                var.set("0")

                    if range_hint:
                        def update_from_scale(val, var=val_var):
```

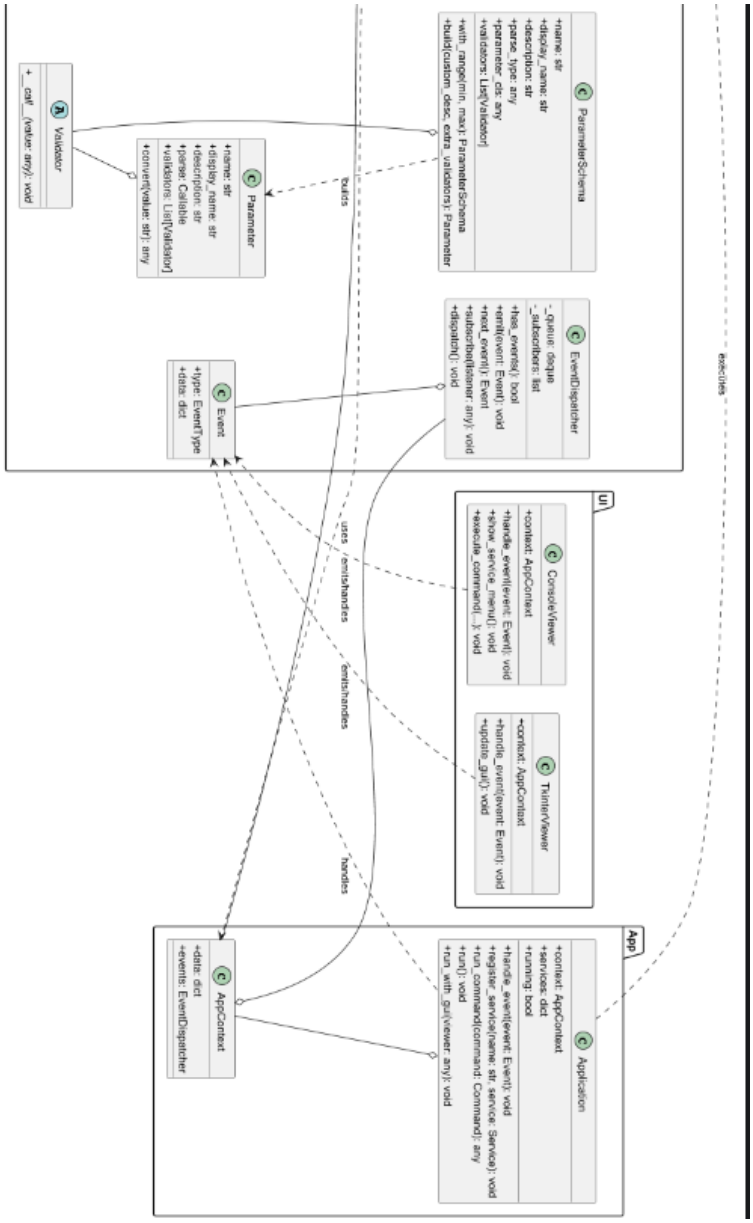
```
                                var.set(str(val))

                            if range_hint.min_value is not None and
range_hint.max_value is not None:
                                entry = tk.Entry(widget_container,
textvariable=val_var, width=10)
                                entry.bind("<FocusOut>", validate_range)
                                entry.pack(side=tk.LEFT, padx=5)

                                scale = tk.Scale(widget_container,
from_=range_hint.min_value, to=range_hint.max_value,
                                                 orient=tk.HORIZONTAL,
length=200, showvalue=0,
                                                 resolution=1 if parse_type
== int else 0.1,
                                                 command=update_from_scale)
                                scale.set(range_hint.min_value)
                                scale.pack(side=tk.LEFT, padx=5)

                                def sync_scale(*args, s=scale, v=val_var,
pt=parse_type):
                                    try:
                                        val = pt(v.get())
                                        s.set(val)
                                    except (ValueError, tk.TclError):
                                        pass

                                val_var.trace_add("write", sync_scale)
                                val_var.set(str(range_hint.min_value))

                            elif range_hint.min_value is not None:
                                spin = tk.Spinbox(widget_container,
from_=range_hint.min_value, to=999999,
                                                  increment=1 if parse_type
== int else 0.1,
                                                  textvariable=val_var,
width=10)
                                spin.bind("<FocusOut>", validate_range)
                                spin.pack(side=tk.LEFT, padx=5)
                                val_var.set(str(range_hint.min_value))
                            elif range_hint.max_value is not None:
                                spin = tk.Spinbox(widget_container, from_=-
999999, to=range_hint.max_value,
                                                  increment=1 if parse_type
== int else 0.1,
                                                  textvariable=val_var,
width=10)
                                spin.bind("<FocusOut>", validate_range)
                                spin.pack(side=tk.LEFT, padx=5)
                                val_var.set(str(range_hint.max_value))
                        else:
                            entry = tk.Entry(widget_container,
textvariable=val_var, width=10)
                            entry.pack(side=tk.LEFT, padx=5)

                    else:
                        val_var = tk.StringVar()
                        self.current_params_values[param_proxy.name] =
val_var
```

```
                    tk.Entry(widget_container, textvariable=val_var,
width=30).pack(side=tk.LEFT)

                error_label = tk.Label(form_frame, text="", fg="red",
bg="white", font=("Arial", 8), relief=tk.RIDGE, bd=1)
                error_label.grid(row=row_index + 1, column=1,
sticky="nsew")
                self.current_params_error_labels[param_proxy.name] =
error_label

            button_container = tk.Frame(self.main_area, bg="white",
height=100)
            button_container.pack(fill=tk.X, side=tk.BOTTOM)

            tk.Button(button_container, text="Виконати", bg="#99ff99",
height=2, width=30,
                      command=lambda:
self.collect_and_execute(command_cls, service,
ui_params)).place(relx=0.5, rely=0.3, anchor=tk.CENTER)
            tk.Button(button_container, text="Скасувати",
command=self.clear_main_area).place(relx=0.5, rely=0.8, anchor=tk.CENTER)

        def collect_and_execute(self, command_cls, service,
params_meta):
            args = []
            is_valid = True

            for label in self.current_params_error_labels.values():
                label.config(text="")

            for p in params_meta:
                try:
                    widget_or_var = self.current_params_values[p.name]
                    if isinstance(widget_or_var, tk.Text):
                        val = widget_or_var.get("1.0", tk.END).strip()
                    elif isinstance(widget_or_var, tk.Listbox):
                        selection = widget_or_var.curselection()
                        if selection:
                            val = widget_or_var.get(selection[0])
                        else:
                            val = ""
                    else:
                        val = widget_or_var.get()

                    converted_val = p.convert(str(val))
                    args.append(converted_val)
                except (ValueError, tk.TclError) as e:
                    is_valid = False
                    error_msg = str(e)
                    if isinstance(e, tk.TclError):
                        error_msg = "Невірний формат числа"

self.current_params_error_labels[p.name].config(text=error_msg)

            if is_valid:
                self.execute_with_params(command_cls, service, args)

        def execute_with_params(self, command_cls, service, args):
            cmd_instance = command_cls(service, *args)
```

```
            self.context.events.emit(Event(EventType.RUN_COMMAND,
command=cmd_instance))

        def update_gui(self):
            try:
                self.root.update()
            except tk.TclError:
                pass
```

Діаграма


Modules
About
GetAboutCommand
+service: AboutService
AboutService
+displayed_name: str
+get_about_creator(): str
+get_about_program(): str
GetAboutCreatorCommand
+_execute(context): str
GetAboutProgramCommand
+_execute(context): str
MetalCalc
CalculateMetalCommand
+service: MetalCalcService
+has_salary: bool
+material_index: int
CalculateMetalRoundCommand
+diameter: float
CalculateElbowCommand
+angle: float
+segments: int
+_execute(context): str
CalculatePipeCommand
+length: float
+_execute(context): str
GetMaterialsCommand
+_execute(context): str
MetalCalcService
+displayed_name: str
+materials: List[MetalMaterial]
+get_materials(): str
+get_materials_count(): int
+get_material_by_id(idx: int): MetalMaterial
+calculate_pipe_unfolding(...)
+calculate_elbow_unfolding(...)
Product
+diameter_mm: float
+material: MetalMaterial
+has_salary: bool
+get_area_m2(): float
+get_salary(): float
+get_cost(): float
«dataclass»
MetalMaterial
+id: int
+name: str
+price_per_m2: float
Pipe
+length_mm: float
Elbow
+angle_deg: float
+segments: int
Core
Service
+context: AppContext
+commands: dict
+register_command(name: str, command_cls: type): void
+get_command_params(command_name: str): List[Parameter]
+emit_event(event: Event): void
Command
+name: str
+description: str
+execute(context: AppContext): void
+_execute(context: AppContext): any
+get_params(service: Service): List[Parameter]
EventType
START
IDLE
STEP
FINISH
UPDATE
ERROR
EXIT
SERVICE_AVAILABLE
SERVICE_SELECTED
COMMAND_AVAILABLE
SHOW_SERVICES_MENU
COMMAND_SELECTED
COMMAND_EXECUTED
EXECUTE_COMMAND
EASTER_EGG
RUN_COMMAND


ParameterSchema
+name: str
+display_name: str
+description: str
+parse_type: any
+parameter_cls: any
+validators: List[Validator]
+with_range(min, max): ParameterSchema
+build(custom_desc, extra_validators): Parameter
EventDispatcher
-_queue: deque
-_subscribers: list
+has_events(): bool
+emit(event: Event): void
+next_event(): Event
+subscribe(listener: any): void
+dispatch(): void
builds
Parameter
+name: str
+display_name: str
+description: str
+parse: Callable
+validators: List[Validator]
+convert(value: str): any
Event
+type: EventType
+data: dict
Validator
+__call__(value: any): void
uses
emits/handles
emits/handles
handles
executes
UI
ConsoleViewer
+context: AppContext
+handle_event(event: Event): void
+show_service_menu(): void
+execute_command(...): void
TkinterViewer
+context: AppContext
+handle_event(event: Event): void
+update_gui(): void
App
Application
+context: AppContext
+services: dict
+running: bool
+handle_event(event: Event): void
+register_service(name: str, service: Service): void
+run_command(command: Command): any
+run(): void
+run_with_gui(viewer: any): void
AppContext
+data: dict
+events: EventDispatcher

Приклад використання модуля

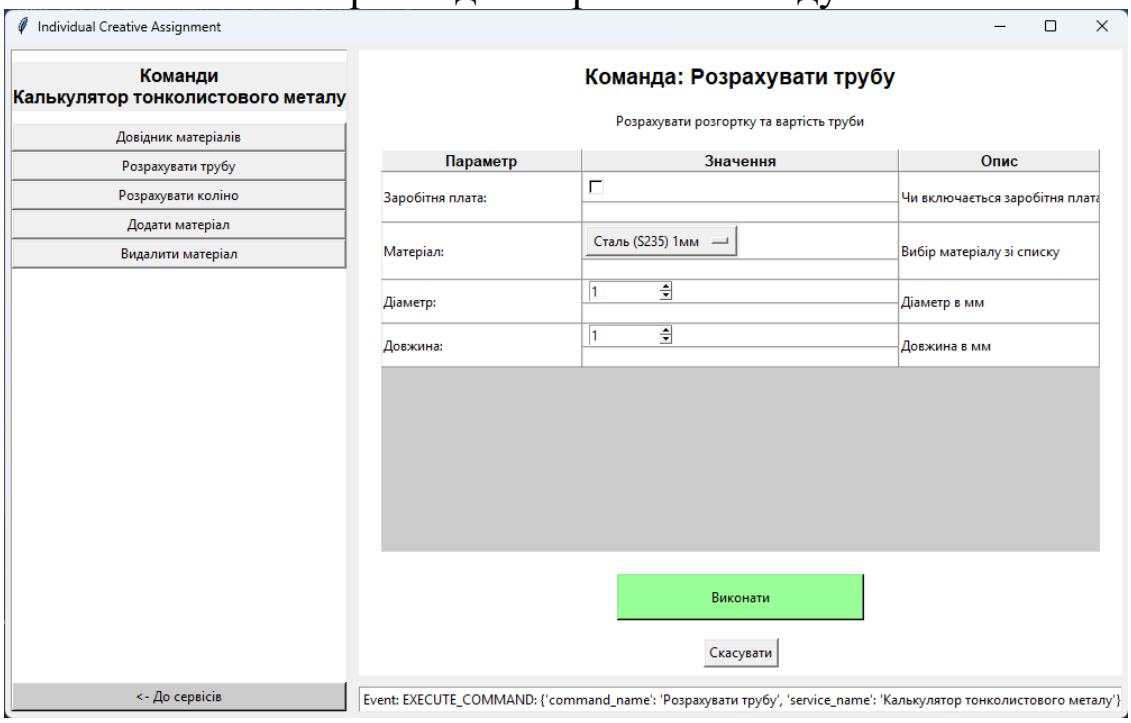


Рисунок 1 - приклад форми команди

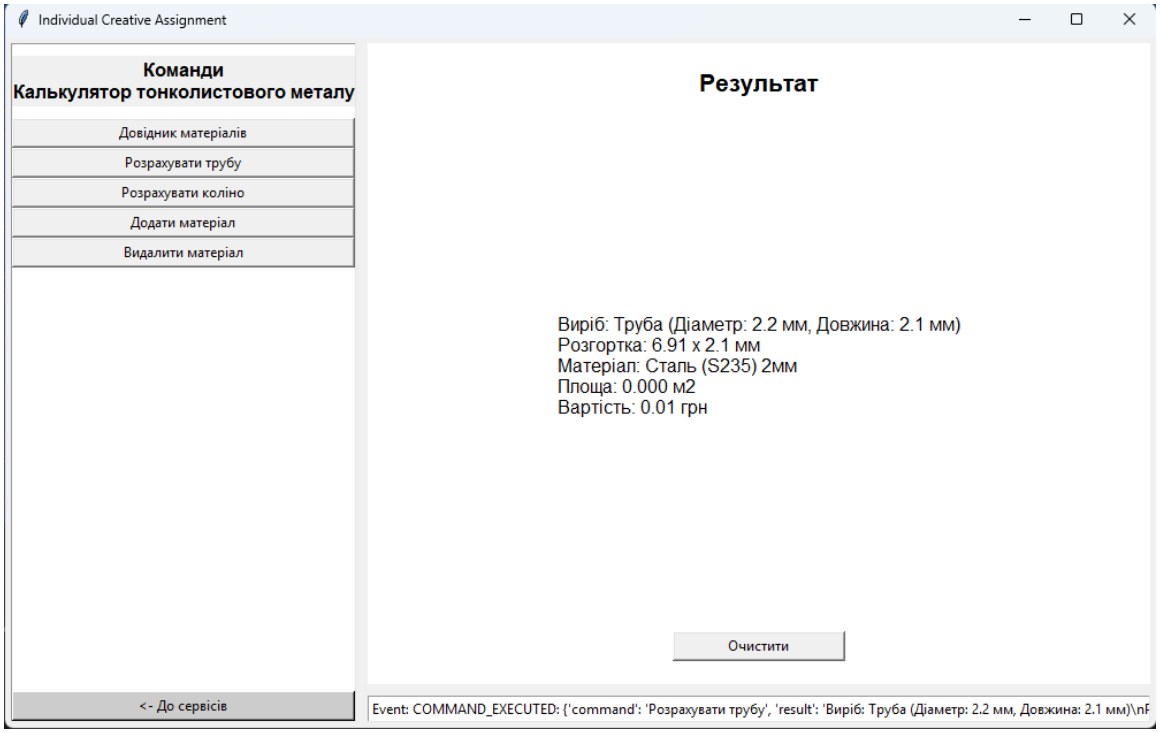


Рисунок 2 - приклад коректного виводу

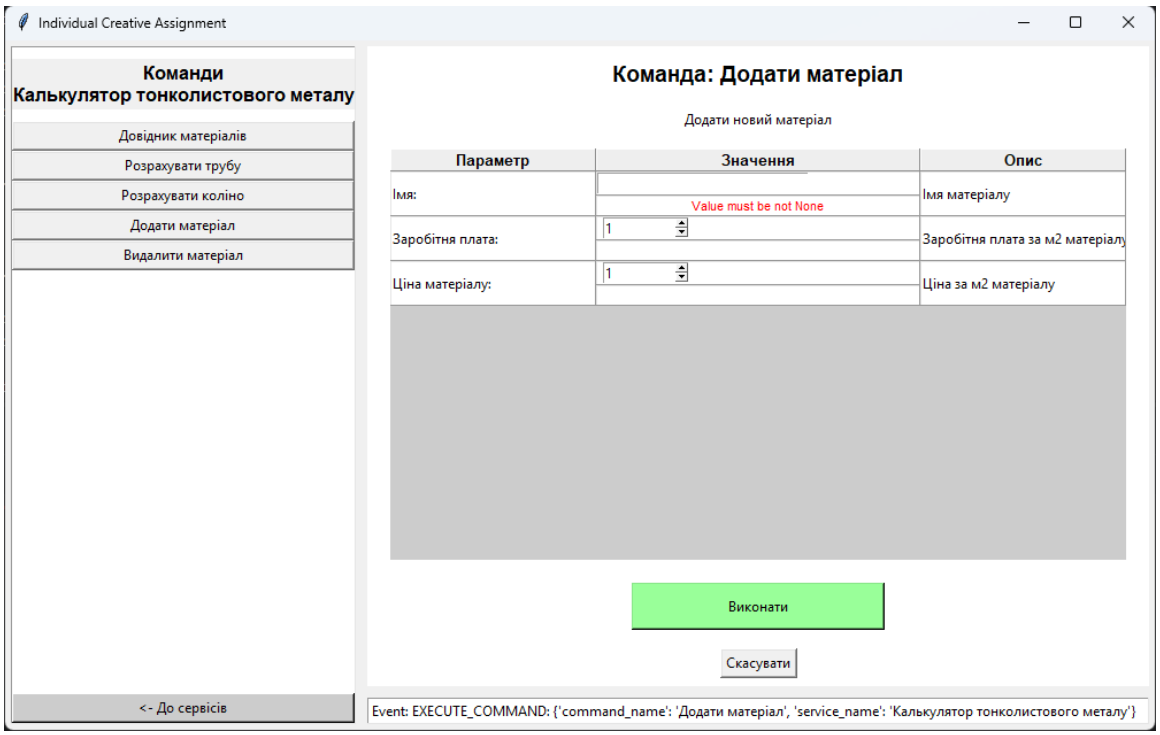


Рисунок 3 - приклад помилки

Висновки

В ході виконання лабораторної було допрацьовано програмний комплекс та розроблено модуль для лабораторної роботи

Джерела

- Посилання на репозиторій https://github.com/Figeiron/IndividualCreativeAssignment/releases/tag/LabWork_13-v0.0.2